

Clearly, $\mathcal{Q}$ still satisfies (i) through (iv), and we claim this collection also satisfies (v). Indeed, let $[a_1, a_2]$ denote any boundary side of $\bigcup_{j=1}^n Q_j$ with its orientation from a_1 to a_2 . By considering the three different possibilities, one sees that a_2 is the beginning point of another boundary side $[a_2, a_3]$. Continuing in this fashion, we obtain a sequence of boundary sides $[a_1, a_2], [a_2, a_3], \dots, [a_n, a_{n+1}], \dots$. Since there are only finitely many sides, we must have $a_n = a_m$ for some n and some $m > n$. We may choose the smallest m so that $a_n = a_m$, say $m = m'$. Then, we note that if $n > 1$, then $a_{m'}$ is an end-point of at least three boundary sides, namely $[a_{n-1}, a_n]$, $[a_n, a_{n+1}]$, and $[a_{m'-1}, a_{m'}]$, hence $a_{m'}$ is a bad boundary vertex. Since we arranged that $\mathcal{Q}$ had no such boundary vertices, we conclude that $n = 1$, and hence the polygon formed by $a_1, \dots, a_{m'}$ is closed and simple. We may repeat this process and find that $\mathcal{Q}$ satisfies property (v), and the proof of Lemma 1.5 is complete.

Finally, we are now able to finish the proof of Theorem 1.2, namely, if Ω is bounded and simply connected we can conclude that Ω^c is connected. To see this, note that if Ω^c is not connected, then we have constructed a curve $\gamma \subset \Omega$ and found a point $w \notin \Omega$ so that $W_\gamma(w) \neq 0$, thus contradicting the fact that Ω is simply connected.

2 The Jordan curve theorem

Although we emphasize in the statement of the theorems which follow that the curves are piecewise-smooth, we note that the proofs involve the use of curves that may only be continuous, (the curves Γ_ϵ below).

The two main results in this section are the following.

Theorem 2.1 *Let Γ be curve in the plane that is simple and piecewise-smooth. Then, the complement of Γ is an open connected set whose boundary is precisely Γ .*

Theorem 2.2 *Let Γ be a curve in the plane which is simple, closed, and piecewise-smooth. Then, the complement of Γ consists of two disjoint connected open sets. Precisely one of these regions is bounded and simply connected; it is called the **interior** of Γ and denoted by Ω . The other component is unbounded, called the **exterior** of Γ , and denoted by $\mathcal{U}$.*

Moreover, with the appropriate orientation for Γ , we have

$$W_\Gamma(z) = \begin{cases} 1 & \text{if } z \in \Omega, \\ 0 & \text{if } z \in \mathcal{U}. \end{cases}$$

Remark. These two theorems continue to hold in the general case where we drop the assumption that the curves are piecewise-smooth. However, as it turns out, the proofs then are more difficult. Fortunately, the restricted setting of piecewise-smooth curves suffices for many applications.

As a consequence of the above propositions, we may state a version of Cauchy's theorem as follows:

Theorem 2.3 *Suppose f is a function that is holomorphic in the interior Ω of a simple closed curve Γ . Then*

$$\int_{\eta} f(\zeta) d\zeta = 0$$

whenever η is any closed curve contained in Ω .

The idea of the proof of Theorem 2.1 can be roughly summarized as follows. Since the complement of Γ is open, it is sufficient to show it is pathwise connected (Exercise 5, Chapter 1). Let z and w belong to the complement of Γ , and join these two points by a curve. If this curve intersects Γ , we first connect z to z' and w to w' , where z' and w' are close to Γ , by curves that do not intersect Γ . Then, we join z' to w' by traveling “parallel” to the curve Γ and going around its end-points if necessary.

Therefore, the key is to construct a family of continuous curves that are “parallel” to Γ . This can be achieved because of the conditions imposed on the curve. Indeed, if γ is a parametrization for a smooth piece of Γ , then γ is continuously differentiable, and $\gamma'(t) \neq 0$. Moreover, the vector $\gamma'(t)$ is tangent to Γ . Consequently, $i\gamma'(t)$ is perpendicular to Γ , and if Γ is simple, considering $\gamma(t) + i\epsilon\gamma'(t)$ amounts to a new curve that is “parallel” to Γ . The details are as follows.

In the next three lemmas and two propositions, we emphasize that Γ_0 denotes a simple *smooth* curve. We recall that an arc-length parametrization γ for a smooth curve Γ_0 satisfies $|\gamma'(t)| = 1$ for all t . Every smooth curve has an arc-length parametrization.

Lemma 2.4 *Let Γ_0 be a simple smooth curve with an arc-length parametrization given by $\gamma : [0, L] \rightarrow \mathbb{C}$. For each real number ϵ , let Γ_{ϵ} be the continuous curve defined by the parametrization*

$$\gamma_{\epsilon}(t) = \gamma(t) + i\epsilon\gamma'(t), \quad \text{for } 0 \leq t \leq L.$$

Then, there exists $\kappa_1 > 0$ so that $\Gamma_0 \cap \Gamma_{\epsilon} = \emptyset$ whenever $0 < |\epsilon| < \kappa_1$.

Proof. We first prove the result locally. If s and t belong to $[0, L]$, then

$$\begin{aligned}\gamma_\epsilon(t) - \gamma(s) &= \gamma(t) - \gamma(s) + i\epsilon\gamma'(t) \\ &= \int_s^t \gamma'(u) du + i\epsilon\gamma'(t) \\ &= \int_s^t [\gamma'(u) - \gamma'(t)] du + (t - s + i\epsilon)\gamma'(t).\end{aligned}$$

Since γ' is uniformly continuous on $[0, L]$, there exists $\delta > 0$ so that $|\gamma'(x) - \gamma'(y)| < 1/2$ whenever $|x - y| < \delta$. In particular, if $|s - t| < \delta$ we find that

$$|\gamma_\epsilon(t) - \gamma(s)| > |t - s + i\epsilon| |\gamma'(t)| - \frac{|t - s|}{2}.$$

Since γ is an arc-length parametrization, we have $|\gamma'(t)| = 1$, and hence

$$|\gamma_\epsilon(t) - \gamma(s)| > |\epsilon|/2,$$

where we have used the simple fact that $2|a + ib| \geq |a| + |b|$ whenever a and b are real. This proves that $\gamma_\epsilon(t) \neq \gamma(s)$ whenever $|t - s| < \delta$ and $\epsilon \neq 0$.

To conclude the proof of the lemma, we argue as follows. (See Figure 2 for an illustration of the argument.)

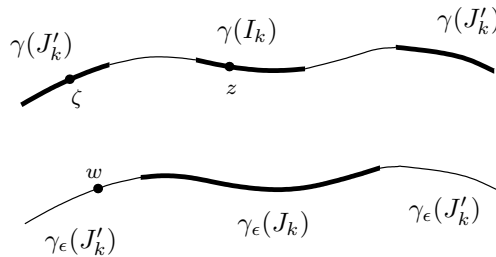


Figure 2. Situation in the proof of Lemma 2.4

Let $0 = t_0 < \dots < t_n = L$ be a partition of $[0, L]$ with $|t_{k+1} - t_k| < \delta$ for all k , and consider

$$I_k = \{t : |t - t_k| \leq \delta/4\}, \quad J_k = \{t : |t - t_k| \leq \delta/2\},$$

and

$$J'_k = \{t : |t - t_k| \geq \delta/2\}.$$

Then, we have just proved that

$$(3) \quad \gamma(I_k) \cap \gamma_\epsilon(J_k) = \emptyset \quad \text{whenever } \epsilon \neq 0.$$

Since Γ_0 is simple, the distance d_k between the two compact sets $\gamma(I_k)$ and $\gamma(J'_k)$ is strictly positive. We now claim that

$$(4) \quad \gamma(I_k) \cap \gamma_\epsilon(J'_k) = \emptyset \quad \text{whenever } |\epsilon| < d_k/2.$$

Indeed, if $z \in \gamma(I_k)$ and $w \in \gamma_\epsilon(J'_k)$, then we choose s in J'_k so that $w = \gamma_\epsilon(s)$ and let $\zeta = \gamma(s)$. The triangle inequality then implies

$$|z - w| \geq |z - \zeta| - |\zeta - w| \geq d_k - |\epsilon| \geq d_k/2,$$

and the claim is established. Finally, if we choose $\kappa_1 = \min_k d_k/2$, then (3) and (4) imply that $\Gamma_0 \cap \Gamma_\epsilon = \emptyset$ whenever $0 < |\epsilon| < \kappa_1$, as desired.

The next lemma shows that any point close to an interior point of the curve belongs to one of its parallel translates. By an interior point of the curve, we mean a point of the form $\gamma(t)$ with t in the open interval $(0, L)$. Such a point should not to be confused with an “interior” point of a curve, as in Theorem 2.2.

Lemma 2.5 *Suppose z is a point which does not belong to the smooth curve Γ_0 , but that is closer to an interior point of the curve than to either of its end-points. Then z belongs to Γ_ϵ for some $\epsilon \neq 0$.*

More precisely, if $z_0 \in \Gamma_0$ is closest to z and $z_0 = \gamma(t_0)$ for some t_0 in the open interval $(0, L)$, then $z = \gamma(t_0) + i\epsilon\gamma'(t_0)$ for some $\epsilon \neq 0$.

Proof. For t in a neighborhood of t_0 the fact that γ is differentiable guarantees that

$$z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|).$$

Since $z_0 = \gamma(t_0)$ minimizes the distance from z to Γ_0 , we find that

$$|z - z_0|^2 \leq |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\operatorname{Re}\left([z - \gamma(t_0)]\overline{\gamma'(t_0)}\right) + o(|t - t_0|).$$

Since $t - t_0$ can take on positive or negative values, we must have $\operatorname{Re}\left([z - \gamma(t_0)]\overline{\gamma'(t_0)}\right) = 0$, otherwise the above inequality can be violated for t close to t_0 . As a result, there exists a real number ϵ with $[z - \gamma(t_0)]\overline{\gamma'(t_0)} = i\epsilon$. Since $|\gamma'(t_0)| = 1$ we have $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, and therefore $z - \gamma(t_0) = i\epsilon\gamma'(t_0)$. The proof of the lemma is complete.

Suppose that z and w are close to interior points of Γ_0 , so that $z \in \Gamma_\epsilon$ and $w \in \Gamma_\eta$ for some non-zero ϵ and η . If ϵ and η have the same sign, we say that the points z and w belong to the same side of Γ_0 . Otherwise, z and w are said to be on opposite sides of Γ_0 . We stress the fact that we do not attempt to define the “two sides of Γ_0 ,” but only that given two points near Γ_0 , we may infer if they are on the “same side” or on “opposite sides”. Also, nothing we have done so far shows that these conditions are mutually exclusive.

Roughly speaking, points on the same side can be joined almost directly by a curve “parallel” to Γ_0 , while for points on opposite sides, we also need to go around one of the end-points of Γ_0 .

We first investigate the situation for points on the same side of Γ_0 .

Proposition 2.6 *Let A and B denote the two end-points of a simple smooth curve Γ_0 , and suppose that K is a compact set that satisfies either*

$$\Gamma_0 \cap K = \emptyset \quad \text{or} \quad \Gamma_0 \cap K = A \cup B.$$

If $z \notin \Gamma_0$ and $w \notin \Gamma_0$ lie on the same side of Γ_0 , and are closer to interior points of Γ_0 than they are to K or to the end-points of Γ_0 , then z and w can be joined by a continuous curve that lies entirely in the complement of $K \cup \Gamma_0$.

The unspecified compact set K will be chosen appropriately in the proof of the Jordan curve theorem.

Proof. By the previous lemma, consider $z_0 = \gamma(t_0)$ and $w_0 = \gamma(s_0)$ that are interior points of Γ_0 closest to z and w , respectively. Then

$$z = \gamma(t_0) + i\epsilon_0\gamma'(t_0) \quad \text{and} \quad w = \gamma(s_0) + i\eta_0\gamma'(s_0),$$

where ϵ_0 and η_0 have the same sign, which we may assume to be positive. We may also assume that $t_0 \leq s_0$.

The hypothesis of the lemma implies that the line segments joining z to z_0 and w to w_0 are entirely contained in the complement of K and Γ_0 . Therefore, for all small $\epsilon > 0$, we may join z and w to the points

$$z_\epsilon = \gamma(t_0) + i\epsilon\gamma'(t_0) \quad \text{and} \quad w = \gamma(s_0) + i\epsilon\gamma'(s_0),$$

respectively. See Figure 3.

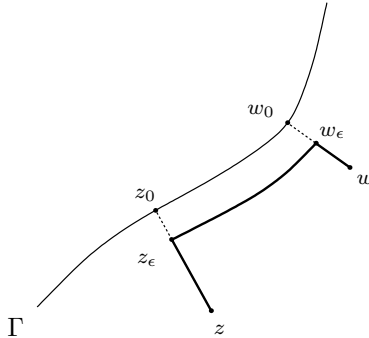


Figure 3. Situation in the proof of Proposition 2.6

Finally, if ϵ is chosen smaller than κ_1 in Lemma 2.4 and also smaller than the distance from K to the part of Γ_0 between z_0 and w_0 , that is, $\{\gamma(t) : t_0 \leq t \leq s_0\}$, then the corresponding part of Γ_ϵ , namely $\{\gamma_\epsilon(t) : t_0 \leq t \leq s_0\}$, joins the point z_ϵ to w_ϵ . Moreover, this curve is contained in the complement of K and Γ_0 . This proves the proposition.

To join points on opposite sides of Γ_0 , we need the following preliminary result, which ensures that there is enough room necessary to travel around the end-points.

Lemma 2.7 *Let Γ_0 be a simple smooth curve. There exists $\kappa_2 > 0$ so that the set N , which consists of points of the form $z = \gamma(L) + \epsilon e^{i\theta}\gamma'(L)$ with $-\pi/2 \leq \theta \leq \pi/2$ and $0 < \epsilon < \kappa_2$, is disjoint from Γ_0 .*

Proof. The argument is similar to the one given in the proof of Lemma 2.4. First, we note that

$$\gamma(L) + \epsilon e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)] du + (L - t + \epsilon e^{i\theta})\gamma'(L).$$

If we choose δ so that $|\gamma'(u) - \gamma'(L)| < 1/2$ when $|u - L| < \delta$, then $|t - L| < \delta$ implies

$$|\gamma(L) + \epsilon e^{i\theta} \gamma'(L) - \gamma(t)| \geq |\epsilon|/2.$$

Therefore $\gamma(t) \notin N$ whenever $L - \delta \leq t \leq L$. Finally, it suffices to choose κ_2 smaller than the distance from the end-point $\gamma(L)$ to the rest of the curve $\gamma(t)$ with $0 \leq t \leq L - \delta$, to conclude the proof.

Finally, we may state the result analogous to Proposition 2.6 for points that could lie on opposite sides of Γ_0 .

Proposition 2.8 *Let A denote an end-point of the simple smooth curve Γ_0 , and suppose that K is a compact set that satisfies either*

$$\Gamma_0 \cap K = \emptyset \quad \text{or} \quad \Gamma_0 \cap K = A.$$

If $z \notin \Gamma_0$ and $w \notin \Gamma_0$ are closer to interior points of Γ_0 than they are to K or to the end-points of Γ_0 , then z and w can be joined by a continuous curve that lies entirely in the complement of $\Gamma_0 \cup K$.

We only provide an outline of the argument, which is similar to the proof of Proposition 2.6. It suffices to consider the case when z and w lie on opposite sides of Γ_0 and $A = \gamma(0)$. First, we may join

$$z_\epsilon = \gamma(t_0) + i\epsilon\gamma'(t_0) \quad \text{and} \quad w_\epsilon = \gamma(s_0) - i\epsilon\gamma'(s_0)$$

to the points

$$z'_\epsilon = \gamma(L) + i\epsilon\gamma'(L) \quad \text{and} \quad w'_\epsilon = \gamma(L) - i\epsilon\gamma'(L).$$

Then, z'_ϵ and w'_ϵ may be joined within the “half-neighborhood” N of Lemma 2.7. Here, if $t_0 \leq s_0$ we must select $|\epsilon|$ smaller than the distance from $\{\gamma(t) : t_0 \leq t \leq L\}$ to K , and also smaller than κ_1 and κ_2 of Lemmas 2.4 and 2.7.

Proof of Theorem 2.1

Let Γ be a simple piecewise-smooth curve.

First, we prove that the boundary of the set $\mathcal{O} = \Gamma^c$ is precisely Γ . Clearly, $\mathcal{O}$ is an open set whose boundary is contained in Γ . Moreover, any point where Γ is smooth also belongs to the boundary of $\mathcal{O}$ (by Lemma 2.4 for instance). Since the boundary of $\mathcal{O}$ must also be closed, we conclude it is equal to all of Γ .

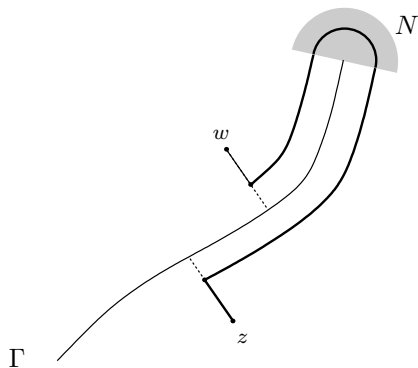


Figure 4. Situation in the proof of Proposition 2.8

The proof that $\mathcal{O}$ is connected is by induction on the number of smooth curves constituting Γ . Suppose first that Γ is simple and smooth, and let Z and W be any two points that do not lie on Γ . Let Λ be any smooth curve in $\mathbb{C}$ that joins Z and W , and which omits the two end-points of Γ . If Λ intersects Γ , it does so at interior points. Therefore, we may join Z by a piece of Λ that does not intersect Γ to a point z that is closer to the interior of Γ than to either of its end-points. Similarly, W can be joined in the complement of Γ to a point w also closer to the interior of Γ than to either of its end-points. Proposition 2.8 (with K empty) then shows that z and w can be joined by a continuous curve in the complement of Γ . Altogether, we may join any two points in the complement of Γ , and this proves the base step of the induction.

Suppose that the theorem is proved for all curves containing $n - 1$ smooth curves, and let Γ consist of n smooth curves, so that we may write

$$\Gamma = K \cup \Gamma_0,$$

where K is the union of $n - 1$ consecutive smooth curves, and Γ_0 is smooth. In particular, K is compact and intersects Γ_0 in a single one of its end-points. By the induction hypothesis, any two points Z and W in the complement of Γ can be joined by a curve that does not intersect K , and we may also assume that this curve omits both end-points of Γ_0 . If this curve intersects Γ_0 in its interior, then we apply Proposition 2.8 to conclude the proof of the theorem.

Proof of Theorem 2.2

Let Γ denote a curve which is simple, closed, and piecewise-smooth. We first prove that the complement of Γ consists of *at most* two components.

Fix a point W that lies outside some large disc that contains Γ , and let $\mathcal{U}$ denote the set of all points that can be joined to W by a continuous curve that lies entirely in the complement of Γ . The set $\mathcal{U}$ is clearly open, and also connected since any two points can be joined by passing first through W . Now we define

$$\Omega = \Gamma^c - \mathcal{U}.$$

We must show that Ω is connected. To this end, let K denote the curve obtained by deleting a smooth piece Γ_0 of Γ . By the Jordan arc theorem, we may join any point $Z \in \Omega$ to W by a curve Λ_Z that does not intersect K . Since $Z \notin \mathcal{U}$, the curve Λ_Z must intersect Γ_0 at one of its interior points. We may therefore choose two points $z, w \in \Lambda_Z$ closer to interior points of Γ_0 than to either of its end-points, and so that the pieces of Λ_Z joining Z to z and W to w are entirely contained in the complement of Γ . Then, the points z and w are on opposite sides of Γ_0 , for otherwise, we could apply Proposition 2.6 to find that Z can be joined to W by a curve lying in the complement of Γ , and this contradicts $Z \notin \mathcal{U}$. Finally, if Z_1 is another point in Ω , the two corresponding points z_1 and w_1 must also lie on opposite sides of Γ_0 . Moreover, z and z_1 must lie on the same side of Γ_0 , for otherwise z and w_1 do, and we can once again join Z to W without crossing Γ , thus contradicting $Z \notin \mathcal{U}$. Therefore, by Proposition 2.6 the points z and z_1 can be joined by a curve in the complement of Γ , and we conclude that Z and Z_1 belong to the same connected component.

The argument thus far proves that Γ^c contains at most two components, but nothing as yet guarantees that Ω is non-empty. To show that Γ^c has precisely two components, it suffices (by Lemma 1.3) to prove that there are points that have different winding numbers with respect to Γ . In fact, we claim that points that are on opposite sides of Γ have winding numbers that differ by 1. To see this, fix a point z_0 on a smooth part of Γ , say $z_0 = \gamma(t_0)$, let $\epsilon > 0$, and define

$$z_\epsilon = \gamma(t_0) + i\epsilon\gamma'(t_0) \quad \text{and} \quad w_\epsilon = \gamma(t_0) - i\epsilon\gamma'(t_0).$$

By our previous observations, points on the same side of Γ belong to the same connected component, and hence

$$\Delta = |W_\Gamma(z_\epsilon) - W_\Gamma(w_\epsilon)|$$

is constant for all small $\epsilon > 0$.

First, we may write

$$\left(\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} \right) = \frac{2i\epsilon\gamma'(t_0)\gamma'(t)}{[\gamma(t) - \gamma(t_0)]^2 + \epsilon^2\gamma'(t_0)^2}.$$

For the numerator, we use

$$\begin{aligned}\gamma'(t) &= \gamma'(t_0) + [\gamma'(t) - \gamma'(t_0)] \\ &= \gamma'(t_0) + \psi(t),\end{aligned}$$

where $\psi(t) \rightarrow 0$ as $t \rightarrow t_0$. For the denominator, we recall that $\gamma'(t_0) \neq 0$, so that

$$[\gamma(t) - \gamma(t_0)]^2 + \epsilon^2\gamma'(t_0)^2 = \gamma'(t_0)^2[(t - t_0)^2 + \epsilon^2] + o(|t - t_0|).$$

Putting these results together, we see that

$$\left(\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} \right) = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2} + E(t),$$

where given $\eta > 0$, there exists $\delta > 0$ so that if $|t - t_0| \leq \delta$, the error term satisfies

$$|E(t)| \leq \eta \frac{\epsilon}{(t - t_0)^2 + \epsilon^2}.$$

We then write

$$\begin{aligned}\Delta &= \frac{1}{2\pi i} \int_{|t - t_0| \geq \delta} \left(\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} \right) dt + \\ &\quad + \frac{1}{2\pi i} \int_{|t - t_0| < \delta} \left(\frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2} + E(t) \right) dt.\end{aligned}$$

The first integral goes to 0 as $\epsilon \rightarrow 0$. In the second integral we make the change of variables $t - t_0 = \epsilon s$, and note that

$$\frac{1}{\pi} \int_{-\rho}^{\rho} \frac{ds}{s^2 + 1} = \frac{1}{\pi} [\arctan s]_{-\rho}^{\rho} \rightarrow 1 \quad \text{as } \rho \rightarrow \infty.$$

We therefore see that letting $\epsilon \rightarrow 0$ gives

$$|\Delta - 1| < \eta.$$

We conclude that $\Delta = 1$, and hence Γ^c has precisely two components. Finally, only one of these components can be unbounded, namely $\mathcal{U}$, and the winding number of Γ in this component must therefore be zero. By our last result, we see that, after possibly reversing the orientation of the curve, the winding number of any point in the bounded component Ω is constant and equal to 1. Also, it is clear from what has been said that any smooth point on Γ can be approached by points in either component, and hence Γ is the boundary of both Ω and $\mathcal{U}$.

The final step in the proof is to show that the interior of the curve, that is, the bounded component Ω , is simply connected. By Theorem 1.2 it suffices to show that Ω^c is connected. If not, then

$$\Omega^c = F_1 \cup F_2,$$

where F_1 and F_2 are closed, disjoint, and non-empty. Let

$$\mathcal{O}_1 = \mathcal{U} \cap F_1 \quad \text{and} \quad \mathcal{O}_2 = \mathcal{U} \cap F_2.$$

Clearly, $\mathcal{O}_1$ and $\mathcal{O}_2$ are disjoint. If $z \in \mathcal{O}_1$, then $z \in \mathcal{U}$, and every small ball centered at z is contained in $\mathcal{U}$. If every such ball intersects F_2 , then $z \in F_2$ since F_2 is closed. However, F_1 and F_2 are disjoint, so this cannot happen. Consequently, $\mathcal{O}_1$ is open, and by the same argument, so is $\mathcal{O}_2$. Finally, we claim that $\mathcal{O}_1$ is non-empty. If not, then F_1 is entirely contained in Γ and $\mathcal{U}$ is contained in F_2 . Pick any point $z \in F_1$, which we know belongs to Γ . Now every ball centered at z intersects $\mathcal{U}$, hence F_2 . But F_2 is closed and disjoint from F_1 , so we get a contradiction. A similar argument for $\mathcal{O}_2$ proves that

$$\mathcal{U} = \mathcal{O}_1 \cup \mathcal{O}_2,$$

where $\mathcal{O}_1, \mathcal{O}_2$ are disjoint, open, and non-empty. This contradicts the fact that $\mathcal{U}$ is connected, and concludes the proof of the Jordan curve theorem for piecewise-smooth curves.

2.1 Proof of a general form of Cauchy's theorem

Theorem 2.9 *If a function f is holomorphic in an open set that contains a simple closed piecewise-smooth curve Γ and its interior, then*

$$\int_{\Gamma} f = 0.$$

Let $\mathcal{O}$ denote an open set on which f is holomorphic, and which contains Γ and its interior Ω . The idea is to construct a closed curve Λ